RACE
Regular Analysis Through Continuous Exercise

Definite Integration: RACE-01

Single Correct Questions :

1. Suppose that $F(x)$ is an antiderivative of $f(x) = \frac{\sin x}{x}, x > 0$ then $\int_1^3 \frac{\sin 2x}{x} dx$ can be expressed as-

(A) $F(6) - F(2)$ (B) $\frac{1}{2}(F(6)-F(2))$ (C) $\frac{1}{2}(F(3)-F(1))$ (D) $2(F(6) - F(2))$

Integer Type Questions :

Evaluate :

2. Let $[t]$ denote the greatest integer $\le t$. Then the value of $8.\int_{-\frac{1}{2}}^{1} ([2x]+|x|)dx$ is _____.

3. $\int_a^b \frac{dx}{\sqrt{1+x^2}}$ where $a = \frac{e-e^{-1}}{2}$ & $b = \frac{e^2-e^{-2}}{2}$

4. If $f(\pi) = 2$ & $\int_0^{\pi} (f(x)+f''(x))\sin x dx = 5$, then find $f(0)$

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Subjective Type Questions:

Evaluate the following definite integrals.

5. **(i)** $\int_0^1 \frac{dx}{x^2+2x\cos\alpha+1}$ where $-\pi < \alpha < \pi$ **(ii)** $\int_0^1 \frac{x^2-1}{1+x^2+x^4}dx$

(iii) $\int_0^{\pi/4} \frac{\sin\theta+\cos\theta}{9+16\sin 2\theta}d\theta$

6. **(i)** $\int_{-1}^1 \frac{x\ dx}{\sqrt{5-4x}}$ **(ii)** $\int_0^{\pi/2} \frac{\cos x\ dx}{(1+\sin x)(2+\sin x)}$

(iii) $\int_0^{\pi/4} \cos 2x\sqrt{1-\sin 2x}\,dx$

7. **(i)** $\int_a^b \frac{|x|}{x}dx$ **(ii)** $\int_1^e \left(\frac{1}{\sqrt{x\ell n x}}+\sqrt{\frac{\ell n x}{x}}\right)dx$ **(iii)** $\int_2^e \left(\frac{1}{\ell n x}-\frac{1}{\ell n^2 x}\right)dx$

8. **(i)** $\int_0^{\pi/2} \frac{dx}{5+4\sin x}$ **(ii)** $\int_2^3 \frac{dx}{(x-1)\sqrt{x^2-2x}}$ **(iii)** $\int_0^{\pi/2} \frac{dx}{1+\cos\theta\,.\cos x}$ $\theta \in (0,\pi)$

9. **(i)** $\int_0^{\pi} \theta\sin^2\theta\cos\theta\,d\theta$ **(ii)** $\int_0^{\pi/2} \frac{x+\sin x}{1+\cos x}dx$ **(iii)** $\int_0^{3\pi/4} ((1+x)\sin x+(1-x)\cos x)dx$

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10. **(i)** $\int_0^{\pi}\left[\cos^2\left(\frac{3\pi}{8}-\frac{x}{4}\right)-\cos^2\left(\frac{11\pi}{8}+\frac{x}{4}\right)\right]dx$

(ii) $\int_0^{\pi/2}\sin\phi\cos\phi\sqrt{\left(a^2\sin^2\phi+b^2\cos^2\phi\right)}\,d\phi\ a\neq b\ (a>0,b>0)$

(iii) $\int_2^3\frac{dx}{\sqrt{(x-1)(5-x)}}$

11. $\int_{\pi/2}^{\pi}x^{\sin x}(1+x\cos x\cdot\ell n\,x+\sin x)dx$

12. If $f(x)=\begin{cases}2 & 0\le x\le 1\\ x+[x] & 1\le x<3\end{cases}$, where [.] denotes the greatest integer function. Evaluate $\int_0^2 f(x)dx$

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Answer Key

1. (A) **2.** (5) **3.** (1) **4.** (3)

5. (i) $\frac{\alpha}{2\sin\alpha}$ *if* $\alpha\neq 0$; $\frac{1}{2}$ *if* $\alpha=0$ (ii) $-\frac{1}{2}\ell n3$ (iii) $\frac{1}{20}\ell n3$

6. (i) $\frac{1}{6}$ (ii) $\ell n\frac{4}{3}$ (iii) $\frac{1}{3}$

7. (i) $|b|-|a|$ (ii) $2\sqrt{e}$ (iii) $e-\frac{2}{\ell n2}$

8. (i) $\frac{2}{3}\tan^{-1}\frac{1}{3}$ (ii) $\frac{\pi}{3}$ (iii) $\frac{\theta}{\sin\theta}$

9. (i) $-\frac{4}{9}$ (ii) $\frac{\pi}{2}$ (iii) $2\left(\sqrt{2}+1\right)$

10. (i) $\sqrt{2}$ (ii) $\left(\frac{a^3-b^3}{a^2-b^2}\right)\frac{1}{3}$ (iii) $\frac{\pi}{6}$ **11.** $\left(\pi-\frac{\pi^2}{4}\right)$ **12.** $\frac{9}{2}$

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Regular Analysis Through Continuous Exercise

Definite Integration: RACE-02

Single Correct Questions :

1. $\int_1^3 \frac{x^2}{x^2-4x+8}dx$ is equal to -

(A) 2 (B) 1 (C) $2\tan^{-1}\frac{1}{2}$ (D) $2\left(1+\tan^{-1}\frac{1}{2}\right)$

2. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{1+\sin^2 x}{1+\pi^{\sin x}}\right)dx$ is

(A) $\frac{\pi}{2}$ (B) $\frac{5\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) $\frac{3\pi}{2}$

Integer Type Questions:

3. Suppose that f, f' and f'' are continuous on $[0, \ell n\ 2]$ and that $f(0)=0, f'(0)=3, f(\ell n\ 2)=6$, $f'(\ell n 2)=4$ and $\int_0^{\ln 2} e^{-2x}\cdot f(x)dx = 3$. Find the value of $\int_0^{\ln 2} e^{-2x}\cdot f''(x)dx$.

4. Suppose that the function f, g, f' and g' are continuous over $[0,1]$, $g(x)\neq 0$ for $x \in [0, 1]$, $f(0)=0, g(0)=\pi, f(1)=\frac{2009}{2}$ and $g(1)=1$.

Find the value of the definite integral, $\int_0^1 \frac{f(x)\cdot g'(x)\{g^2(x)-1\}+f'(x)\cdot g(x)\{g^2(x)+1\}}{g^2(x)}dx$.

Subjective Type Questions:

Evaluate :

5. Evaluate : $\int_{-\sqrt{2}}^{\sqrt{2}} \frac{2x^7+3x^6-10x^5-7x^3-12x^2+x+1}{x^2+2}\,dx$

6. Evaluate : $\int_{-1}^{1}\frac{(2x^{332}+x^{998}+4x^{1668}.\sin x^{691})}{1+x^{666}}dx$

7. **(i)** $\int_0^3 |x^2-x-2|dx$ **(ii)** $\int_0^{\pi/2}|\sin x-\cos x|dx$ **(iii)** $\int_0^{\pi/2}\sqrt{\frac{1-\sin 2x}{1+\sin 2x}}dx$

8. $\int_0^1 x\ f''(x)\,dx$, where $f(x)=\cos(\tan^{-1}x)$

9. Given $f'(x)=\frac{\cos x}{x}$, $f\left(\frac{\pi}{2}\right)=a$, $f\left(\frac{3\pi}{2}\right)=b$. Find the value of the definite integral $\int_{\pi/2}^{3\pi/2} f(x)dx$.

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10. **(i)** $\int_{-1}^{1}\left(\frac{d}{dx}\left(\frac{1}{1+e^{1/x}}\right)\right)dx$

(ii) $\int_{\ell n\,2}^{\ell n\,3} f(x)dx$, where $f(x)=e^{-x}+2e^{-2x}+3e^{-3x}+\ldots\infty$

11. $\int_{\pi/6}^{\pi/3}\frac{dx}{1+\tan^5 x}$

12. $\int_{0}^{3}\sqrt{\frac{x}{3-x}}dx$

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Answer Key

1.	(A)	**2.**	(C)	**3.**	(13)	**4.**	(2009)
5.	$\frac{\pi}{2\sqrt{2}}-\frac{16\sqrt{2}}{5}$	**6.**	$\frac{\pi+4}{666}$				
7.	(i) $\frac{31}{6}$	(ii)	$2(\sqrt{2}-1)$	(iii)	ln2	**8.**	$1-\frac{3}{2\sqrt{2}}$
9.	$2-\frac{\pi}{2}(a-3b)$	**10.**	(i) $\frac{2}{1+e}$	(ii)	$\frac{1}{2}$	**11.**	$\frac{\pi}{12}$
12.	$\frac{3\pi}{2}$						

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Regular Analysis Through Continuous Exercise

Definite Integration: RACE-03

Single Correct Questions :

1. $\int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx$ is equal to :

(A) 6 (B) 8 (C) 5 (D) 10

2. $\int_0^{\pi/2n} \frac{dx}{1 + \tan^n nx} =$

(A) 0 (B) $\frac{\pi}{4n}$ (C) $\frac{n\pi}{4}$ (D) $\frac{\pi}{2n}$

3. If $\int_0^{\pi} \frac{x^3 \cos^4 x \sin^2 x dx}{(\pi^2 - 3\pi x + 3x^2)} = k \int_0^{\pi/2} \sin^2 x dx$, then the value of k is :-

(A) $\frac{\pi}{12}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{16}$ (D) $\frac{\pi}{8}$

4. If $g(x) = \int_0^x \cos^4 t\, dt$, then $g(x + \pi)$ equals

(A) $g(x) + g(\pi)$ (B) $g(x) - g(\pi)$ (C) $g(x)\, g(\pi)$ (D) $[g(x)/g(\pi)]$

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5. The value of $\int_{-1}^{1} x^2 e^{[x^3]} dx$, where [t] denotes the greatest integer $\le t$, is :

(A) $\frac{e-1}{3e}$ (B) $\frac{e+1}{3}$ (C) $\frac{e+1}{3e}$ (D) $\frac{1}{3e}$

Integer Type Questions

6. The value of $I = \left| \int_{-3}^{3} \ell n \left(\frac{(\sqrt{1+x+x^2})(x-[x])}{(\sqrt{1-x+x^2})} \right) dx \right|$ is (where [.] denotes greatest integer function)

Subjective Type Questions :

7. $\int_{1/3}^{3} \frac{\sin^{-1} \frac{x}{\sqrt{1+x^2}}}{x} dx$

8. **(i)** $\int_0^{\pi} \frac{(ax+b)\sec x \tan x}{4 + \tan^2 x} dx (a, b > 0)$

(ii) $\int_0^{\pi} \frac{(2x+3)\sin x}{(1+\cos^2 x)} dx$

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9. Prove that :

(a) $\int_{\alpha}^{\beta}\sqrt{\frac{x-\alpha}{\beta-x}}dx=(\beta-\alpha)\frac{\pi}{2}$

(b) $\int_{\alpha}^{\beta}\frac{x\,.\,dx}{\sqrt{(x-\alpha)(\beta-x)}}=(\alpha+\beta)\frac{\pi}{2}=$, where $\alpha<\beta$

10. $\int_{0}^{\pi/2}\ell n\left(\sin^2 x\cos x\right)dx$

11. $\int_{-\pi/2}^{\pi/2}\sqrt{\cos x-\cos^3 x}\,dx$

12. $\int_{-1.5}^{10}\{2x\}dx$, where {.} denotes fractional part of x.

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Answer Key

Question	1	2	3	4	5	6
Answer	C	B	D	A	C	6

7. $\frac{\pi\ell n3}{2}$ **8.** (i) $\frac{(a\pi+2b)\pi}{3\sqrt{3}}$ (ii) $\frac{\pi(\pi+3)}{2}$

10. $-\left(\frac{3\pi}{2}\right)\ell n2$ **11.** $\frac{4}{3}$ **12.** $\frac{23}{4}$

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Regular Analysis Through Continuous Exercise

Definite Integration: RACE-04

Single Correct Questions :

1. If the value of the integral $\int_0^5 \frac{x+[x]}{e^{x-[x]}}dx = \alpha e^{-1} + \beta$, where $\alpha, \beta \in \mathbb{Q}$, $5\alpha + 6\beta = 0$, and $[x]$ denotes the greatest integer less than or equal to x; then the value of $(\alpha + \beta)^2$ is equal to :

(A) 100 (B) 25 (C) 16 (D) 36

2. If $f(x)$ is an even function which is also periodic with the period T and $\int_0^a f(x)dx = 3$ and $\int_{-T/2}^{3T/2} f(x)dx = 18$, then $\int_{-a}^{a+5T} f(x)dx$ is equal to -

(A) 96 (B) 93 (C) 51 (D) 48

3. Let $f:(a,b) \to R$ be twice differentiable function such that $f(x) = \int_a^x g(t)dt$ for a differentiable function $g(x)$. If $f(x) = 0$ has exactly five distinct roots in (a,b), then $g(x)g'(x) = 0$ has at least :

(A) twelve roots in (a,b) (B) five roots in (a,b)
(C) seven roots in (a,b) (D) three roots in (a,b)

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4. Let $y = f(x)$ be a differentiable curve satisfying $\int_2^x f(t)dt = \frac{x^2}{2} + \int_x^2 t^2 f(t)dt$, then $\int_{-\pi/4}^{\pi/4} \frac{f(x)+x^9-x^3+x+1}{\cos^2 x}dx$ equals -

(A) 0 (B) 1 (C) 2 (D) 4

5. The value of $\lim_{n\to\infty} \sum_{r=0}^{2n-1} \frac{n}{n^2+4r^2}$ is :

(A) $\frac{1}{2}\tan^{-1}(2)$ (B) $\frac{1}{2}\tan^{-1}(4)$ (C) $\tan^{-1}(4)$ (D) $\frac{1}{4}\tan^{-1}(4)$

6. If $L = \lim_{n\to\infty}\left(\frac{n+2}{n^2+n+1} + \frac{n+4}{n^2+2n+4} + \frac{n+6}{n^2+3n+9} + \ldots\ldots + \frac{5}{7n}\right)$, then e^L is equal to -

(A) 1 (B) $\frac{1}{7}$ (C) 7 (D) e

Integer Type Question :

7. If $\lambda = \lim_{n\to\infty} \frac{\left(\sum_{r=1}^n r\right)\left(\sum_{r=1}^n r^2\right)\left(\sum_{r=1}^n r^3\right)}{\sum_{r=1}^n r^8}$, then the value of 8λ is -

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8. If $\int_0^{\pi/4} \sin^6 x\cos^4 x dx = \frac{3\pi}{1024} - \frac{K}{960}$, then K is equal to

Subjective Type Questions:

9. $\int_{20\pi+\frac{\pi}{6}}^{20\pi+\frac{\pi}{3}} (\sin x + \cos x)dx$

10. If $f(x) = \int_{1/x}^{\sqrt{x}} \sin t\, dt$, then find $f'(1)$.

11. Find the constants 'a' ($a > 0$) and 'b' such that $\lim_{x\to 0} \frac{\int_0^x \frac{t^2\, dt}{\sqrt{a+t}}}{bx - \sin x} = 1$.

12. Prove the inequalities :

(i) $\frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \frac{\pi\sqrt{2}}{8}$

(ii) $2e^{-1/4} < \int_0^2 e^{x^2-x} dx < 2e^2$

(iii) $\frac{1}{3} < \int_0^1 x^{(\sin x+\cos x)^2} dx < \frac{1}{2}$

(iv) $\frac{1}{2} \le \int_0^2 \frac{dx}{2+x^2} \le \frac{5}{6}$

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Answer Key

Question	1	2	3	4	5	6	7	8
Answer	B	C	C	C	B	C	3	3

9. $\left(\sqrt{3}-1\right)$ **10.** $\frac{3}{2}\sin 1$ **11.** $a = 4$ and $b = 1$

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